

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

ME-801(A) (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Energy Management And Audit
(Elective - III)
Time : Three Hours

Maximum Marks : 70

- Note :** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain for the Energy policy:
 - i) Purpose
 - ii) Prospective contents
 - iii) Formulations and
 - iv) Rectification
2. List and enumerate the Goals of Energy Audit and where they can be applied.
OR
Explain the different types of representations in Energy consumption.
3. Discuss important tips for energy conservation in the following sectors.
 - i) Lighting and Refrigeration
 - ii) Transportation
4. a) Discuss the areas of losses in electrical motors.
b) Define heat exchanger. State its types with neat sketches.
5. a) What is Energy balance diagram? Explain its objectives.
b) What are key elements of energy action planning?
6. The integrated paper plant has produced 119366 MT of paper during the year 2014-15. The management has implemented various energy conservation measures as part of PAT scheme and reduced the specific energy consumption from 53GJ/ tonne of product to 50 GJ/tonne of product. The actual production during the assessment year (2016-17) is 124141 MT. Calculate the plant energy performance and state your inference.

[2]

7. a) Compare:
- i) Micro energy audit
 - ii) Macro audit
 - iii) Billing audit
 - iv) Field audit
- b) State the duties and role of the energy auditor in an industry.
8. Answer any four of the following:
- a) Assume yourself as energy manager, what step you would take for energy management for your organization.
 - b) Nowadays the concept of optimization in input energy requirement has evolved by substitution in fuel and energy. Explain.
 - c) Discuss the Sankey diagram and its use in material energy balance.
 - d) Discuss various methods for 'Building energy management'.
 - e) Explain thermal insulation in thermal systems. How thickness is determined?
 - f) How heat can be recovered from waste?
